

```
##pip install y-data profiling
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('/content/Titanic-Dataset.csv')
```

```
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	892	0	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN	Q
1	893	1	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	NaN	S
2	894	0	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN	Q
3	895	0	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN	S
4	896	1	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN	S

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
# pip install pandas-profiling --user
# pip install ydata-profiling --user
```

◆ Gemini

```
!pip install ydata-profiling
from ydata_profiling import ProfileReport

Requirement already satisfied: typeguard<5,>=4 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (4.5.1)
Collecting imagehash==4.3.2 (from ydata-profiling)
  Downloading ImageHash-4.3.2-py2.py3-none-any.whl.metadata (8.4 kB)
Requirement already satisfied: wordcloud>=1.9.4 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (1.9.6)
Collecting dacite<2,>=1.9 (from ydata-profiling)
  Downloading dacite-1.9.2-py3-none-any.whl.metadata (17 kB)
Requirement already satisfied: numba<0.63,>=0.60 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (0.60.0)
Requirement already satisfied: PyWavelets in /usr/local/lib/python3.12/dist-packages (from imagehash==4.3.2->ydata-profiling) (1.9.0)
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packages (from imagehash==4.3.2->ydata-profiling) (11.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2<3.2,>=3.1.6->ydata-profiling) (3.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (0.1)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (0.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (4.22.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (1.3.1)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (20.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (3.0)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (2.7)
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba<0.63,>=0.60->ydata-profiling) (0.43.0dev0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas!=1.4.0,<3.0,>1.5->ydata-profiling) (2020.1)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas!=1.4.0,<3.0,>1.5->ydata-profiling) (2022.7)
Requirement already satisfied: joblib>=0.14.1 in /usr/local/lib/python3.12/dist-packages (from phik<0.13,>=0.12.5->ydata-profiling) (1.5)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=2->ydata-profiling) (0.6.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=2->ydata-profiling) (2.41.4)
Requirement already satisfied: typing-extensions>=4.14.1 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=2->ydata-profiling) (4.14.1)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=2->ydata-profiling) (0.4.2)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.32.0->ydata-profiling) (3.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.32.0->ydata-profiling) (3.13)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.32.0->ydata-profiling) (2.32.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.32.0->ydata-profiling) (2025.11.12)
Requirement already satisfied: patsy>=0.5.6 in /usr/local/lib/python3.12/dist-packages (from statsmodels<1,>=0.13.2->ydata-profiling) (1.0)
```

import ydata_profiling is deprecated and will not receive more updates.

Please install fg-data-profiling via `pip install fg-data-profiling` and use `import data\_profiling` instead.

from ydata_profiling import ProfileReport

prof= ProfileReport(df)
prof.to_file(output_file='output.html')

Summarize dataset: 100%46/46 [00:07<00:00, 5.71it/s, Completed]

0%|██████████| 0/12 [00:00<?, ?it/s]
25%|███████| 3/12 [00:00<00:00, 20.74it/s]
50%|██████████| 6/12 [00:00<00:00, 24.40it/s]
100%|██████████| 12/12 [00:00<00:00, 29.18it/s]

Generate report structure: 100%1/1 [00:04<00:00, 4.54s/it]

Render HTML: 100%1/1 [00:00<00:00, 1.75it/s]

Export report to file: 100%1/1 [00:00<00:00, 33.74it/s]

df.shape

(418, 12)

df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 12 columns):
Column Non-Null Count Dtype
--- --
0 PassengerId 418 non-null int64
1 Survived 418 non-null int64
2 Pclass 418 non-null int64
3 Name 418 non-null object
4 Sex 418 non-null object
5 Age 418 non-null float64
6 SibSp 418 non-null int64
7 Parch 418 non-null int64
8 Ticket 418 non-null object
9 Fare 417 non-null float64
10 Cabin 418 non-null object
11 Embarked 418 non-null object
dtypes: float64(2), int64(5), object(5)
memory usage: 39.3+ KB

df.describe()

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare	
count	418.000000	418.000000	418.000000	418.000000	418.000000	418.000000	417.000000	
mean	1100.500000	0.363636	2.265550	29.599282	0.447368	0.392344	35.627188	
std	120.810458	0.481622	0.841838	12.703770	0.896760	0.981429	55.907576	
min	892.000000	0.000000	1.000000	0.170000	0.000000	0.000000	0.000000	
25%	996.250000	0.000000	1.000000	23.000000	0.000000	0.000000	7.895800	
50%	1100.500000	0.000000	3.000000	27.000000	0.000000	0.000000	14.454200	
75%	1204.750000	1.000000	3.000000	35.750000	1.000000	0.000000	31.500000	
max	1309.000000	1.000000	3.000000	76.000000	8.000000	9.000000	512.329200	

df['Age'].describe()

Age

count 418.000000
mean 29.599282
std 12.703770
min 0.170000
25% 23.000000
50% 27.000000
75% 35.750000
max 76.000000

dtype: float64

df.isnull().sum()

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	1
Cabin	0
Embarked	0

dtype: int64

df['Age'].unique()

array([34.5 , 47. , 62. , 27. , 22. , 14. , 30. , 26. , 18. ,
21. , 46. , 23. , 63. , 24. , 35. , 45. , 55. , 9. ,
48. , 50. , 22.5 , 41. , 33. , 18.5 , 25. , 39. , 60. ,
36. , 20. , 28. , 10. , 17. , 32. , 13. , 31. , 29. ,
28.5 , 32.5 , 6. , 67. , 49. , 2. , 76. , 43. , 16. ,
1. , 12. , 42. , 53. , 26.5 , 40. , 61. , 60.5 , 7. ,
15. , 54. , 64. , 37. , 34. , 11.5 , 8. , 0.33, 38. ,
57. , 40.5 , 0.92, 19. , 36.5 , 0.75, 0.83, 58. , 0.17,
59. , 14.5 , 44. , 5. , 51. , 3. , 38.5])

df['Age'] = df['Age'].fillna(df.Age.median())

df['Age'].unique()

array([34.5 , 47. , 62. , 27. , 22. , 14. , 30. , 26. , 18. ,
21. , 46. , 23. , 63. , 24. , 35. , 45. , 55. , 9. ,
48. , 50. , 22.5 , 41. , 33. , 18.5 , 25. , 39. , 60. ,
36. , 20. , 28. , 10. , 17. , 32. , 13. , 31. , 29. ,
28.5 , 32.5 , 6. , 67. , 49. , 2. , 76. , 43. , 16. ,
1. , 12. , 42. , 53. , 26.5 , 40. , 61. , 60.5 , 7. ,
15. , 54. , 64. , 37. , 34. , 11.5 , 8. , 0.33, 38. ,
57. , 40.5 , 0.92, 19. , 36.5 , 0.75, 0.83, 58. , 0.17,
59. , 14.5 , 44. , 5. , 51. , 3. , 38.5])

df.Embarked.unique()[2]

'C'

df['Embarked'] = df['Embarked'].fillna(df.Embarked.mode()[0])

df.Embarked.unique()

array(['Q', 'S', 'C'], dtype=object)

df.Cabin.unique()

array(['B57 B59 B63 B66', 'B45', 'E31', 'B36', 'A21', 'C78', 'D34', 'D19',
'A9', 'D15', 'C31', 'C23 C25 C27', 'F G63', 'B61', 'C53', 'D43',
'C130', 'C132', 'C101', 'C55 C57', 'B71', 'C46', 'C116', 'F',
'A29', 'G6', 'C6', 'C28', 'C51', 'E46', 'C54', 'C97', 'D22', 'B10',
'F4', 'E45', 'E52', 'D30', 'B58 B60', 'E34', 'C62 C64', 'A11',
'B11', 'C80', 'F33', 'C85', 'D37', 'C86', 'D21', 'C89', 'F E46',
'A34', 'D', 'B26', 'C22 C26', 'B69', 'C32', 'B78', 'F E57', 'F2',
'A18', 'C106', 'B51 B53 B55', 'D10 D12', 'E60', 'E50', 'E39 E41',
'B52 B54 B56', 'C39', 'B24', 'D28', 'B41', 'C7', 'D40', 'D38',
'C105'], dtype=object)

df.isnull().sum()

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	1
Cabin	0
Embarked	0

dtype: int64

```
df['Cabin'] = df['Cabin'].fillna(df.Cabin.mode()[0])
```

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	1
Cabin	0
Embarked	0

dtype: int64

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  418 non-null    int64
1   Survived     418 non-null    int64
2   Pclass       418 non-null    int64
3   Name         418 non-null    object
4   Sex          418 non-null    object
5   Age          418 non-null    float64
6   SibSp        418 non-null    int64
7   Parch        418 non-null    int64
8   Ticket       418 non-null    object
9   Fare         417 non-null    float64
10  Cabin        418 non-null    object
11  Embarked     418 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 39.3+ KB
```

```
df.columns
```

```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
      'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='object')
```

```
X =df.drop(columns=['PassengerId', 'Survived','Name','Fare','Ticket','Cabin'])
y=df['Survived']
```

X

	Pclass	Sex	Age	SibSp	Parch	Embarked	
0	3	male	34.5	0	0	Q	
1	3	female	47.0	1	0	S	
2	2	male	62.0	0	0	Q	
3	3	male	27.0	0	0	S	
4	3	female	22.0	1	1	S	
...	...	...	...	...	...	...	
413	3	male	27.0	0	0	S	
414	1	female	39.0	0	0	C	
415	3	male	38.5	0	0	S	
416	3	male	27.0	0	0	S	
417	3	male	27.0	1	1	C	

418 rows × 6 columns

Next steps:

[Generate code with X](#)

[New interactive sheet](#)

y

	Survived
0	0
1	1
2	0
3	0
4	1
...	...
413	0
414	1
415	0
416	0
417	0

418 rows × 1 columns

dtype: int64

```
# Encoding (ordinal encoding , label encoding, onehot encoding)
# from sklearn.preprocessing import OrdinalEncoder
# Get dummies from pandas Library
pd.get_dummies(X,columns=['Sex','Pclass'],dtype=int)
```

	Age	SibSp	Parch	Embarked	Sex_female	Sex_male	Pclass_1	Pclass_2	Pclass_3	
0	34.5	0	0	Q	0	1	0	0	1	
1	47.0	1	0	S	1	0	0	0	1	
2	62.0	0	0	Q	0	1	0	1	0	
3	27.0	0	0	S	0	1	0	0	1	
4	22.0	1	1	S	1	0	0	0	1	
...	...	...	...	...	...	...	...	...	...	
413	27.0	0	0	S	0	1	0	0	1	
414	39.0	0	0	C	1	0	1	0	0	
415	38.5	0	0	S	0	1	0	0	1	
416	27.0	0	0	S	0	1	0	0	1	
417	27.0	1	1	C	0	1	0	0	1	

418 rows × 9 columns

```
# drop first
X=pd.get_dummies(X,columns=['Sex','Pclass','Embarked'],drop_first=True,dtype=int)
```

X

	Age	SibSp	Parch	Sex_male	Pclass_2	Pclass_3	Embarked_Q	Embarked_S	
0	34.5	0	0	1	0	1	1	0	
1	47.0	1	0	0	0	1	0	1	
2	62.0	0	0	1	1	0	1	0	
3	27.0	0	0	1	0	1	0	1	
4	22.0	1	1	0	0	1	0	1	
...	...	...	...	...	...	...	...	...	
413	27.0	0	0	1	0	1	0	1	
414	39.0	0	0	0	0	0	0	0	
415	38.5	0	0	1	0	1	0	1	
416	27.0	0	0	1	0	1	0	1	
417	27.0	1	1	1	0	1	0	0	

418 rows × 8 columns

Next steps:

[Generate code with X](#)

[New interactive sheet](#)

X.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Age         418 non-null    float64
1   SibSp       418 non-null    int64
2   Parch       418 non-null    int64
3   Sex_male    418 non-null    int64
4   Pclass_2    418 non-null    int64
5   Pclass_3    418 non-null    int64
6   Embarked_Q  418 non-null    int64
7   Embarked_S  418 non-null    int64
dtypes: float64(1), int64(7)
memory usage: 26.3 KB
```

```
#pip install --upgrade scikit-learn
```

```
from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2,random_state=90)
```

X_train

	Age	SibSp	Parch	Sex_male	Pclass_2	Pclass_3	Embarked_Q	Embarked_S	
368	45.0	0	1	0	0	0	0	0	
153	36.0	0	2	0	0	1	0	1	
346	26.0	0	0	1	1	0	0	1	
20	55.0	1	0	1	0	0	0	0	
25	50.0	1	0	1	0	1	0	1	
...	...	...	...	...	...	...	...	...	
323	33.0	0	0	1	0	0	0	1	
159	26.0	0	2	0	0	1	0	1	
114	63.0	1	0	0	0	0	0	1	
157	23.0	0	0	0	0	1	0	1	
347	38.0	0	0	0	0	1	0	0	

334 rows × 8 columns

Next steps:

[Generate code with X_train](#)

[New interactive sheet](#)

y_test

Survived

221	0
310	0
130	0
273	1
103	0
...	...
261	0
65	1
344	0
119	1
43	1

84 rows × 1 columns

dtype: int64

```
from sklearn.neighbors import KNeighborsClassifier
```

```
knn = KNeighborsClassifier(n_neighbors=3)
```

```
knn.fit(X_train,y_train)#/200
```

```
▼ KNeighborsClassifier ⓘ ?  
KNeighborsClassifier(n_neighbors=3)
```

```
y_pred=knn.predict(X_test)
```

```
pd.DataFrame({'Ytest':y_test,'Ypred':y_pred})
```

	Ytest	Ypred	
221	0	0	
310	0	0	
130	0	0	
273	1	1	
103	0	0	
...	...	...	
261	0	0	
65	1	1	
344	0	0	
119	1	1	
43	1	1	

84 rows × 2 columns

```
from sklearn.metrics import accuracy_score  
accuracy_score(y_test, y_pred)
```

0.8095238095238095

```
from sklearn.metrics import accuracy_score, recall_score, precision_score, f1_score, confusion_matrix, classification_report  
print('Accuracy Score: ', accuracy_score(y_test, y_pred))  
print('Recall Score: ', recall_score(y_test, y_pred))  
print('Precision Score: ', precision_score(y_test, y_pred))  
print('F1 Score: ', f1_score(y_test, y_pred))  
print('Confusion Matrix: \n', confusion_matrix(y_test, y_pred))  
print('Classification Report: \n', classification_report(y_test, y_pred))
```

```
Accuracy Score: 0.8095238095238095  
Recall Score: 0.6818181818181818  
Precision Score: 0.625  
F1 Score: 0.6521739130434783  
Confusion Matrix:  
[[53  9]  
 [ 7 15]]  
Classification Report:
```

	precision	recall	f1-score	support
0	0.88	0.85	0.87	62
1	0.62	0.68	0.65	22
accuracy			0.81	84
macro avg	0.75	0.77	0.76	84
weighted avg	0.82	0.81	0.81	84

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